Probability with Engineering Applications: Week12

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Programme

- Mean squared error
- Minimum mean square error estimators
- Binary hypothesis testing

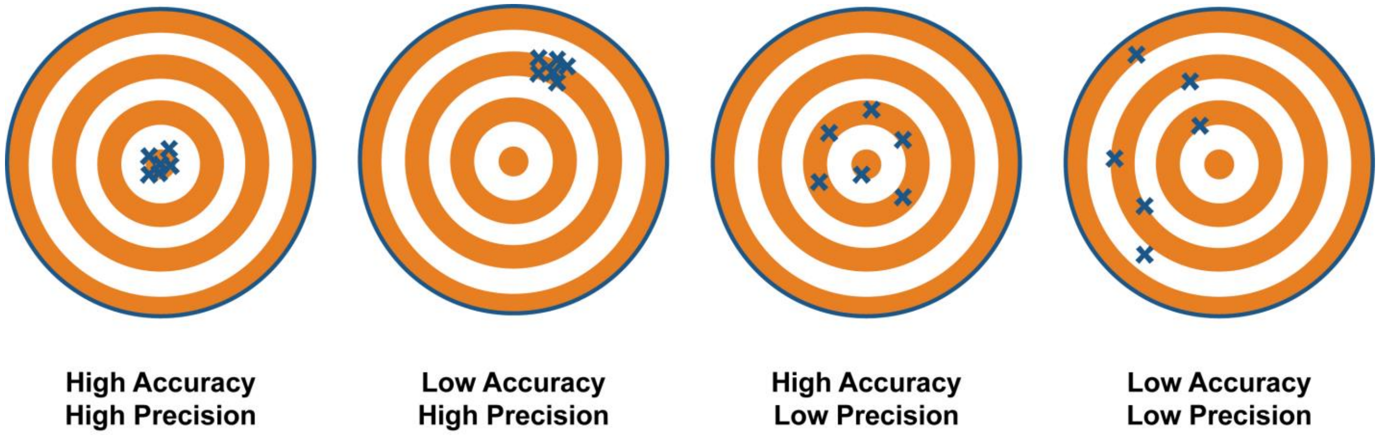
Efficiency and mean squared error

Bias and variance

If is (often) desirable to seek unbiased estimators, i.e., to find estimators $\hat{\mu}$ and $\widehat{\sigma^2}$ such that

$$E[\mu - \hat{\mu}] = 0 \quad \text{and} \quad E\left[\sigma^2 - \widehat{\sigma^2}\right] = 0$$

Ideal estimators have high accuracy (unbiased) and high precision (low variance)



Drawing balls from an urn

An urn contains N balls, labeled $\{1, 2, \ldots, N\}$. We draw three balls, with replacement.

How can we estimate N, based on this sample x_1, x_2, x_3?

$$T_1 = X_1 \qquad E[T_1] = \frac{1}{2}(N+1)$$

$$T_2 = \overline{X}_3 \qquad E[T_2] = \frac{1}{2}(N+1)$$

$$T_3 = \max(X_1, X_2, X_3) \qquad E[T_3] = \frac{3}{4}N + \frac{1}{2} - \frac{1}{4N}$$

Mean squared error

Definition:
Let T be an estimator for the parameter θ.
The <u>mean squared error</u> of T is defined as:

$$\mathrm{MSE}(T) = \mathrm{E}[(T - \theta)^2].$$

Corollary:

$$\mathrm{MSE}(T) = \mathrm{Var}(T) + (\mathrm{E}[T] - \theta)^2$$

$$= \mathrm{Variance} + \mathrm{Bias}^2$$

Efficiency

Definition:
Let T_1 and T_2 be two estimators for the same parameter. If

$$\mathrm{MSE}(T_2) < \mathrm{MSE}(T_1),$$

we say that T_2 is <u>more efficient</u> than T_1.

Corollary:
Let T_1 and T_2 be unbiased estimators for the same parameter. If

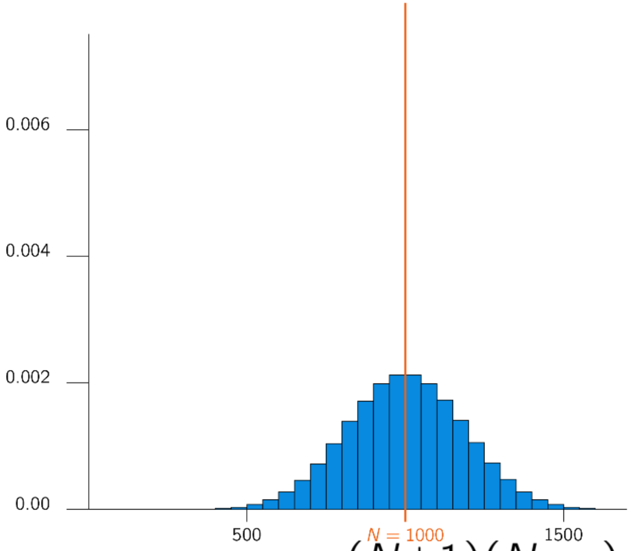
$$\mathrm{Var}(T_2) < \mathrm{Var}(T_1),$$

then T_2 is more efficient than T_1.

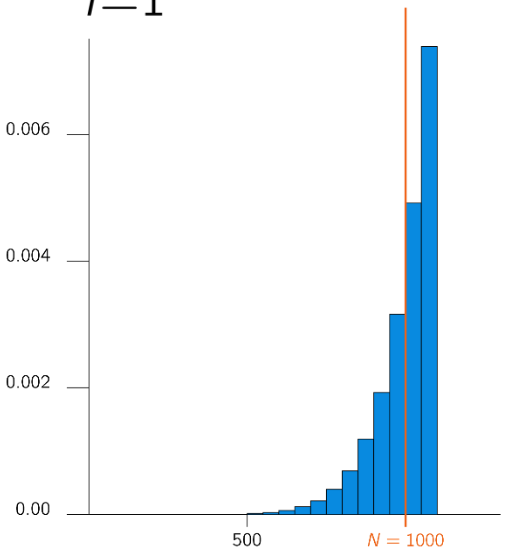
Urn without replacement (German Tanks)

The ball numbers X_i of draws from an urn without replacement are modeled as random variables from $\{1, 2, \ldots, N\}$ with uniform distribution. Two unbiased estimators for N are

$$T_1 = 2\bar{X}_n - 1 \quad \text{and} \quad T_2 = \frac{n+1}{n} \max_{i=1}^{n}\{X_i\} - 1.$$



$$\mathrm{Var}(T_1) = \frac{(N+1)(N-n)}{3n}$$

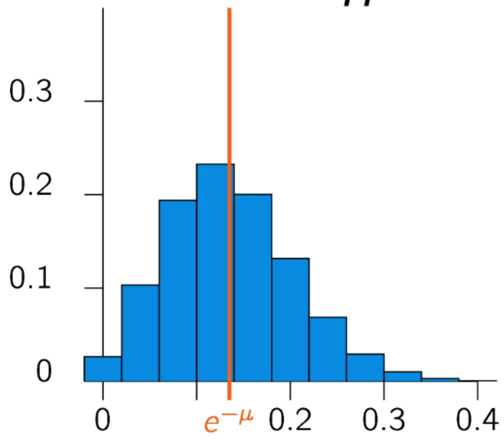


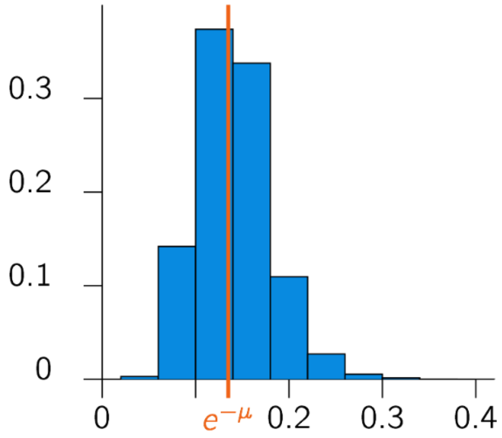
$$\mathrm{Var}(T_2) = \frac{(N+1)(N-n)}{(n+2)n}$$

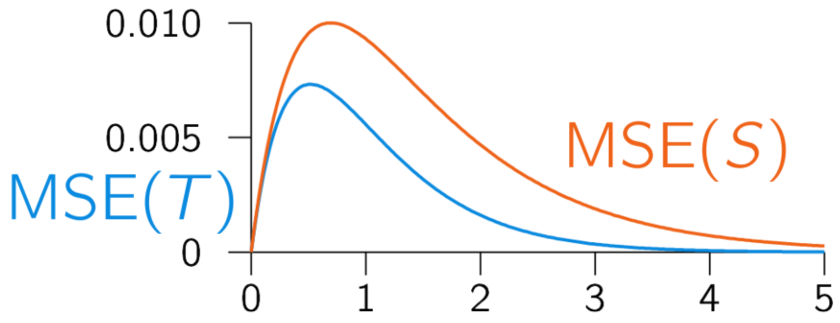
Downtime on a network server

The number of packets arriving at a server in one minute is modeled as $X_i \sim \text{Pois}(\mu)$. Two estimators for $P(X_i = 0) = e^{-\mu}$ are

$$S = \frac{\text{number of } X_i = 0}{n} \quad \text{and} \quad T = e^{-\bar{X}_n}$$







Estimators and efficiency

Consider the following two statements about two estimators S and T:

1. If $\text{Var}(S) < \text{Var}(T)$, then S is more efficient than T.
2. If S is unbiased and T is biased, then always $\text{Var}(S) < \text{Var}(T)$.

A. Only 1 is true

B. Only 2 is true

C. 1 and 2 are both false

D. 1 and 2 are both true

Minimum mean square error estimators

MMSE Estimation by a Constant

Suppose we wish to estimate a random variable Y with known distribution by a constant δ

We will consider the more general cases soon

The estimation error for estimating Y by a constant δ is the random variable

$$e = Y - \delta$$

The mean square error (MSE) if Y is continuous-type is given by

$$\mathrm{MSE} = E[(Y - \delta)^2] = \int_{-\infty}^{\infty} (v - \delta)^2 \, f_Y(v) \, dv$$

MMSE Estimation by a Constant

Another way to write the MSE is with

$$\begin{aligned}\text{MSE} &= E[Y^2 - 2\delta Y + \delta^2] \\ &= E[Y^2] - 2\delta E[Y] + \delta^2\end{aligned}$$

Now, differentiate and equate to 0:

$$\frac{d\text{MSE}}{d\delta} = -2E[Y] + 2\delta^* = 0 \quad \Rightarrow \delta^* = E[Y]$$

The minimum mean square error (MMSE) estimator of Y by a constant δ is given by

$\delta^* = E[Y]$

The MSE in this case is?

MMSE Estimation: General Case

Goal: Estimate Y based on an observable random variable X when the joint pdf (or pmf) of X and Y is known

Objective: Determine the function g such that the estimator $g(X)$ minimizes the mean square error (MSE):

$$g^*(X) = \underset{g}{\operatorname{argmin}}\, E([Y - g(X)]^2)$$

The random variable $g^*(X)$ is called the unconstrained MMSE optimal estimator of Y given X

MMSE Estimator: General Case

Recall: The conditional expectation of Y given $X = u$ is given by

$$E[Y|X = u] = \int_{-\infty}^{\infty} v\, f_{Y|X}(v|u)\, dv$$

which is a function of u.

MMSE Estimator: General Case

Stated in words: the MMSE estimator of Y given X is the expected value of Y conditioned on X.

Is the MMSE estimator unbiased?

Linear MMSE Estimators

It is often impossible to determine the MMSE estimator

- $E[Y|X=u] = \int_{-\infty}^{\infty} v\, f_{Y|X}(v|u)\, dv$ may not have a closed form expression
- It may require massive computation to calculate $E[Y|X=u]$
- The joint pdf $f_{X,Y}$ may be unknown

Let us consider linear minimum mean square error (LMMSE) estimators of Y given X

Objective: Determine the function $L(X) = aX + b$ (i.e., the constants a and b) that minmizes

$$\mathrm{MSE} = E[(Y - [aX + b])^2]$$

LMMSE Estimators

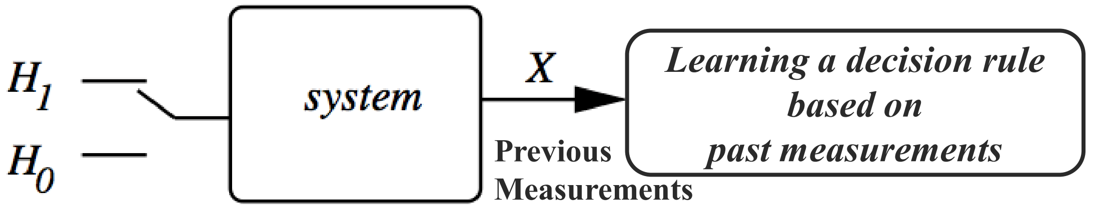
Prop. The LMMSE estimator of Y given X is

$$\hat{E}[Y|X] = \mu_Y + \left[\frac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)}\right](X - \mu_X)$$

$$= \mu_Y + \sigma_Y \rho_{Y,X} \left(\frac{X - \mu_X}{\sigma_X}\right)$$

Binary hypothesis testing

Binary Hypothesis Testing:
Basic Framework



MRI Scan Machine

H1: Patient has a tumor
H0: Patient doesn't have a tumor

Images captured from brain when H1 or H0:
e.g. X = No. of abnormal spots in the image

Binary: Either hypothesis H_1 is true or hypothesis H_0 is true.

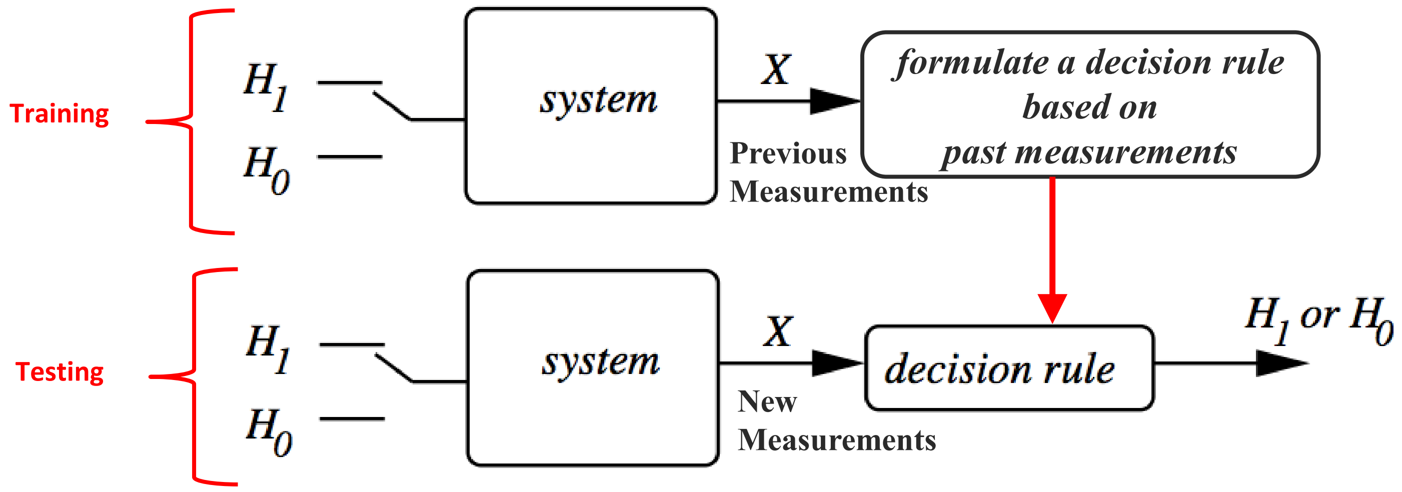
Based on which hypothesis is true, system generates a set of observations X.

We use the past observations and find **the conditional probabilities** of different patterns under each hypothesis:

– e.g. What is the probability of observing two abnormal spots (X = 2) given patient has a tumor (H1)?

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.0	0.1	0.3	0.6
H_0	0.4	0.3	0.2	0.1

Binary Hypothesis Testing:
Basic Framework



Either hypothesis H_1 is true or hypothesis H_0 is true.

Based on which hypothesis is true, system generates a set of observations *X*.

We measure the past observations to **learn the conditional probabilities** of different patterns under each hypothesis:

- e.g. What is the probability of observing two abnormal spots (X = 2) if patient has a tumor (H1)?

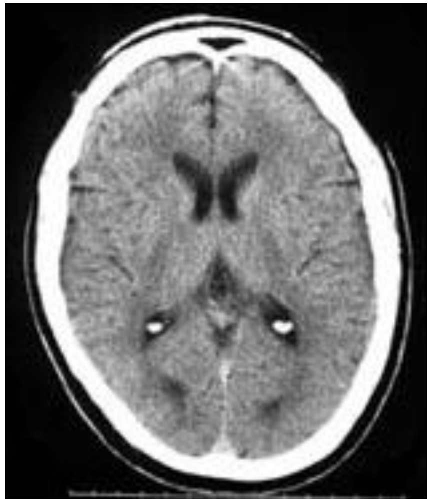
Next we **formulate a decision rule** that is then used to determine whether H_1 or H_0 is true for **new measurements**.

Note that the system has uncertainty or the features are not the best, so the decision rule can sometimes declare a true hypothesis, or it can make an error

Example 1

Suppose the data is from a computer aided tomography (CAT) scan system, and the hypotheses are:

- ***H_1***: A tumor is present
- ***H_0***: No tumor is present



We model the number of abnormal spots observed in the image by a discrete random variable *X.*

Suppose:

- If hypothesis H_1 is true (given), then X has pmf p_1
- If hypothesis H_0 is true (given), then X has pmf p_0

Note that p1 and p0 are conditional pmf's, described by a ***likelihood Matrix***

		$X=0$	$X=1$	$X=2$	$X=3$
P (X = k | H1) =>	H_1	0.0	0.1	0.3	0.6
P (X = k | H0) =>	H_0	0.4	0.3	0.2	0.1

Given a tumor, what is the probability of observing 3 spots

Example 1: Decision Rule

A decision rule specifies for each possible observation (each of the possible values of *X*), which hypothesis is declared.

Typically a decision rule is described using a likelihood matrix. Here we underline one entry in each column, to specify which hypothesis is to be declared for each possible value of *X.*

Example: A *probability-based* decision rule: H1 is declared when the probability of an observation given hypothesis H1 is higher than its probability given H0:

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.0	0.1	$\underline{0.3}$	$\underline{0.6}$
H_0	$\underline{0.4}$	$\underline{0.3}$	0.2	0.1
	Declare H0		Declare H1	

If we observe X = 2,3 in the new measurements, H1 (tumor) is declared, otherwise H0 (no tumor).

Example *intuitive* decision rule: H_1 is declared whenever $X \geq 1$:

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.0	$\underline{0.1}$	$\underline{0.3}$	$\underline{0.6}$
H_0	$\underline{0.4}$	0.3	0.2	0.1

$\leftarrow$ underlines indicate the decision rule used for this example.

Binary Hypothesis Testing

In a binary hypothesis testing problem, there are two possibilities for which the hypothesis is true, and two possibilities for which hypothesis is declared:

H_0 = Negative or null hypothesis,

H_1 = Positive or alternative hypothesis

So there are four possible outcomes:

1. **H_1 is declared given Hypothesis H_1 is true. => True Positive**
2. **H_1 is declared given Hypothesis H_0 is true. => False Positive (False Alarm)**
3. **H_0 is declared given Hypothesis H_0 is true. => True Negative**
4. **H_0 is declared given Hypothesis H_1 is true. => False Negative (Miss Detection)**

Probability of False Alarm and Miss

Two conditional probabilities are defined:

$$\boxed{p_{false-alarm} = P(declare\ H_1\ true \mid H_0)}$$ => **Type I Error**

$$\boxed{p_{miss} = P(declare\ H_0\ true \mid H_1)}$$ => **Type II Error**

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.0	$\underline{0.1}$	$\underline{0.3}$	$\underline{0.6}$
H_0	$\underline{0.4}$	0.3	0.2	0.1

Decision Rule

If we modify the sample decision rule in the previous example to declare H_1 when $X = 1$:

So what decision rule should be used?

- Trade-off between the two types of error probabilities
- Evaluate the pair of conditional error probabilities $(p_{false-alarm}, p_{miss})$ for multiple decision rules and then make a final selection

Maximum Likelihood (ML) Decision Rule

Maximum likelihood (ML) decision rule declares the hypothesis that ***maximizes the probability (likelihood) of the observation***

$$P(X = k \mid H_0) \quad and \quad P(X = k \mid H_1)$$

ML decision rule is based on ***likelihood matrix,*** the matrix of conditional probabilities of $P(X = k \mid H_i)$

ML decision rule is specified by underlining the larger entry in each column of likelihood matrix.

If the entries in a column are identical, then either can be underlined.

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.0	0.1	$\underline{0.3}$	$\underline{0.6}$
H_0	$\underline{0.4}$	$\underline{0.3}$	0.2	0.1

Maximum a posteriori probability (MAP) Decision Rule

Maximum a-posteriori probability (MAP) decision rule declares the hypothesis which ***maximizes the posteriori probabilities***

A Posteriori probability is a conditional probability that an observer would declare a hypothesis, given an observation *k:*

$$P(H_0 \mid X = k) \quad and \quad P(H_1 \mid X = k)$$

So given an observation $X = k$, the MAP decision rule chooses the hypothesis with the larger posteriori probability

Maximum a posteriori probability (MAP) Decision Rule (Cont'd)

So the MAP decision rule requires the computation of the ***joint probabilities***:

But we have:

- $P(X = k \mid H_i)$ from the likelihood matrix
- $P(H_0)$ *and* $P(H_1)$ which are called the ***prior probabilities***.

The prior probabilities are determined independent of the experiment and prior to any observation is made.

Maximum a posteriori probability (MAP) Decision Rule (Cont'd)

MAP decision rule creates a ***joint probability matrix:*** this is a matrix of the joint probabilities of $P(H_i, X = k)$.

Each row H_i in the joint probability matrix is π_i times the corresponding row of the likelihood matrix.

The sum of entries in row H_i is π_i and the some of all the entries in the matrix is one.

$$\text{Likelihood Matrix} \xrightarrow{\pi_0=0.8\quad,\quad \pi_1=0.2} \text{Joint Probability Matrix}$$

Likelihood Matrix:

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.0	0.1	0.3	0.6
H_0	0.4	0.3	0.2	0.1

Joint Probability Matrix:

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.00	0.02	0.06	0.12
H_0	0.32	0.24	0.16	0.08.

Maximum a posteriori probability (MAP) Decision Rule (Cont'd)

MAP decision rule is specified by underlying the larger entry in each column of the joint probability matrix.

If the entries in a column are identical, then either can be underlined.

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.00	0.02	0.06	$\underline{0.12}$
H_0	$\underline{0.32}$	$\underline{0.24}$	$\underline{0.16}$	0.08

Average Error Probability

Remember the conditional probabilities of:

$$\boxed{p_{false-alarm} = P(declare\ H_1\ true \mid H_0)}$$

$$\boxed{p_{miss} = P(declare\ H_0\ true \mid H_1)}$$

The **average error probability** is defined as:

$$\boxed{p_e = \pi_0 p_{false-alarm} + \pi_1 p_{miss}}$$

p_e is the sum of all the entries in the *joint probability* matrix that are not underlined.

Among all decision rules, MAP is the one that minimizes p_e =>Optimal

	$X=0$	$X=1$	$X=2$	$X=3$
H_1	0.00	0.02	0.06	$\underline{0.12}$
H_0	$\underline{0.32}$	$\underline{0.24}$	$\underline{0.16}$	0.08

Likelihood Ratio Test (LRT)

A way to generalize the ML and MAP decision rules into a single framework is using LRT.

Define the likelihood ratio $\Lambda(k)$ for each possible observation *k* as the ratio of the two conditional probabilities:

$$\Lambda(k) = \frac{p_1(k)}{p_0(k)} = \frac{P(X = k \mid H_1)}{P(X = k \mid H_0)}$$

A decision rule can be expressed as an LRT with threshold τ:

$$\Lambda(X) \begin{cases} > \tau & \textit{declare } H_1 \textit{ is true.} \\ < \tau & \textit{declare } H_0 \textit{ is true.} \end{cases}$$

What happens to $p_{false-alarm}$ and p_{miss} if τ increases?

Likelihood Ratio Test (LRT) (Cont'd)

If the observation is *X = k:*

The ML rule declares hypothesis H_1 is true, if $p_1(k) > p_0(k)$ and otherwise it declares H_0 is true.

So the ML rule can be specified using an LRT with $\tau = 1$:

$$\Lambda(X) = \frac{p_1(k)}{p_0(k)} = \begin{cases} >1 & \textit{declare } H_1 \textit{ is true.} \\ <1 & \textit{declare } H_0 \textit{ is true.} \end{cases}$$

The MAP rule declares hypothesis H_1 is true, if $\pi_1 p_1(k) > \pi_0 p_0(k)$ and otherwise it declares H_0 is true.

So the MAP rule can be specified using an LRT with $\tau = \frac{\pi_0}{\pi_1}$:

$$\Lambda(X) = \frac{p_1(k)}{p_0(k)} = \begin{cases} > \dfrac{\pi_0}{\pi_1} & \textit{declare } H_1 \textit{ is true.} \\ < \dfrac{\pi_0}{\pi_1} & \textit{declare } H_0 \textit{ is true.} \end{cases}$$

For uniform priors $\pi_0 = \pi_1$, MAP and ML decision rules are the same

Example 2: Discrete Case

Suppose you have a coin and you know that either:

- H1: the coin is biased, showing heads on each flip with probability 2/3; or
- H0: the coin is fair, showing heads and tails with probability 1/2

Suppose you flip the coin five times. Let *X* be the number of times heads shows.

Describe the ML and MAP decision rules using LRT.

Find $p_{false-alarm}$, p_{miss}, and p_e for both decision rules.

Use the prior probabilities of $(\pi_0, \pi_1) = (0.2, 0.8)$

Example 2 (Cont'd)

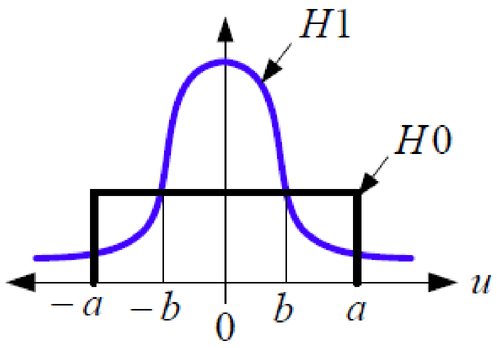
The rows of the likelihood matrix consist of the pmf of *X:*

	X = 0	*X* = 1	*X* = 2	*X* = 3	*X* = 4	*X* = 5
H_1	$\left(\frac{1}{3}\right)^5$	$5\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^4$	$10\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^3$	$10\left(\frac{2}{3}\right)^3\left(\frac{1}{3}\right)^2$	$5\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)$	$\left(\frac{2}{3}\right)^5$
H_0	$\left(\frac{1}{2}\right)^5$	$5\left(\frac{1}{2}\right)^5$	$10\left(\frac{1}{2}\right)^5$	$10\left(\frac{1}{2}\right)^5$	$5\left(\frac{1}{2}\right)^5$	$\left(\frac{1}{2}\right)^5$

Example 3

An observation X is drawn from a standard normal distribution (i.e. N(0,1)) if hypothesis H1 is true and from a uniform distribution with support [-a, a], if hypothesis H0 is true.

As shown in the figure, the pdfs of the two distributions are equal when |u| = b.



a) Describe the maximum likelihood (ML) decision rule in terms of observation X and constant values *a* and *b*.

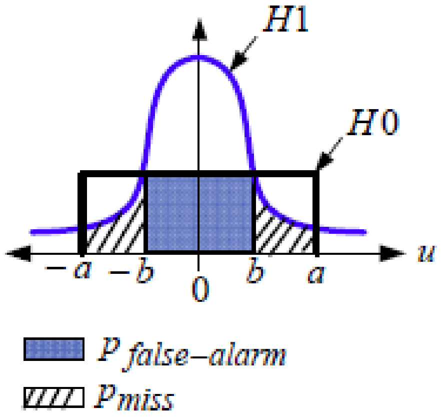
b) Shade and label the regions in the figure such that the area of one region is the $p_{false-alarm}$ and the area of the other region is the p_{miss}.

c) Express the $p_{false-alarm}$ and p_{miss} for ML decision rule in terms of *a* and *b* and the $\Phi(u)$, i.e., the CDF of the standard normal distribution.

Example 3 (Cont'd)

(a) From the figure, the pdf for H_1 is smaller than the pdf for H_0 precisely when $b < |u| < a$. Thus, the ML rule is given by:

$$\hat{H} = \begin{cases} H_0 & b < |X| < a \\ H_1 & \text{otherwise} \end{cases}$$

(b) 

(c) These probabilities are calculated as follows:

$$\begin{aligned} p_{false\ alarm} &= \frac{2b}{2a} = \frac{b}{a} \\ p_{miss} &= 2(\Phi(a) - \Phi(b)) = 2(Q(b) - Q(a)) \end{aligned}$$